Analysis Data Reviewer's Guide (ADRG)

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Field	Value
Document	Analysis Data Reviewer's Guide (ADRG)
Study	TROPIC / EFC6193 / NCT00417079
Compound	Cabazitaxel (CbzP) vs Mitoxantrone (MP)
Standards	CDISC ADaMIG v1.3 OCCDS v1.0 (+ custom episode-merge; not "OCCDS v1.1")
Document version	1.3 (audit closure / portfolio release)
Effective	2026-08-05
Supersedes	ADRG v1.2 from the v0.2.0-portfolio release (which superseded the v0.1.0-demo-rc.1 baseline)
Product claim	Path A only -- controlled non-submission demonstration

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0. What this package is / is not (read first)

This package is	This package is not
SDTM -> ADaM -> TFL -> Define -> eCTD-style programming demonstration	An FDA filing / NDA package
Dual-language (SAS 9.4 / R) recon on real MP ADaM under a genuine SAS engine when seals say oda/local	Organizational GxP two-programmer double programming
Risk-tiered admiral third engine for ADSL + OS/PFS	Full ADaM re-derived in admiral
Controlled TFL catalog (21 in-scope IDs; 18 SAP IDs deferred)	Full SAP Appendix D TFL inventory
Hash-sealed Path A demo RC	21 CFR Part 11 validated system
Comparative TFLs that may include synthetic/reconstructed CbzP	Independent clinical confirmation of CbzP efficacy

Binding claim: docs/PRODUCT_CLAIM.md

Residual risks (ACCEPTED findings): docs/workstreams/WS5_KNOWN_DIFFERENCES_MEMO.md

Current portfolio release: tag v0.2.2-portfolio docs/RELEASE_NOTE_v0.2.2-portfolio.md python3 scripts/verify_release.py

Review package face: 08_submission_package/m5/ 08_submission_package/README.md

SAP v4.0 lock: 02_specifications/sap/TROPIC_SAP_v4.0_industry_grade.docx is the programming authority for this demonstration (lock memo: 06_qc_evidence/audit/SAP_LOCK_REVIEW_MEMO.md). It is not a sponsor-approved filing SAP. This ADRG explains implementation; it does not invent new analysis decisions.

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1. Study & demonstration overview

The TROPIC Phase III Trial (NCT00417079) evaluated cabazitaxel (25 mg/m2 IV q3w) + prednisone vs mitoxantrone (12 mg/m2 IV q3w) + prednisone in mCRPC after docetaxel (de Bono et al., Lancet 2010).

This repository is a Path A programming demonstration built on:

- Real MP arm (N=371) de-identified SDTM (Sanofi 2013 / Project Data Sphere) -> dual-language ADaM + recon
- Synthetic / reconstructed CbzP (N=378) merged only at TFL for comparative displays (non-confirmatory; F-003)

In the published trial, cabazitaxel carried a substantial safety burden (~82% Grade 3/4 neutropenia; ~8% febrile neutropenia). Synthetic lab outputs in this package approximate published rates for pipeline exercise only (see T-21); they are not a re-analysis of trial safety.

Comparative Optimus-style exposure-response displays (e.g. F-17) exercise dose-toxicity methods on disclosed synthetic/reconstructed comparator content. They must not be cited as independent clinical dose-optimization evidence.

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2. Key Derivations & Episode Merging

Myeloid/Neutropenic Episode Merging (ADAE)

Under the published CDISC OCCDS v1.0 structure, separate adverse event records (e.g. repeated reports of neutropenia) artificially inflate the event count denominator. We therefore apply a pre-specified custom 3-day continuous episode-merging extension (this merging rule is a TROPIC analysis convention -- it is not part of the published OCCDS v1.0; no CDISC "OCCDS v1.1" standard exists):

1. Within a patient and Customized Query 02 (CQ02NAM = 'HEMATOLOGIC irAE'), neutropenic events with a start date within 3 days of the previous event's end date are merged.
2. The continuous start (CIAESDT), end (CIAEEDT), and duration (CIAEDUR) are calculated across the merged sequence.
3. The occurrence flag AEOCCFL is set to 'Y' only for the first record in the merged sequence, establishing an accurate, non-inflated safety denominator.

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3. Project Optimus Modeling Parameters (ADLB)

To support dose-toxicity modeling, two continuous parameters were derived per cycle:

- ANCNADIR (PARAMCD: ANCNADIR): The absolute minimum ANC value recorded during the primary nadir window (Day 4 to Day 24).
- ANCRECDY (PARAMCD: ANCRECDY): The number of days from the nadir date to the first post-nadir assessment where ANC >= 1.5 x103/uL, defining the patient-specific recovery latency.
- Exposure Linkage: The internally consistent subject-level source RDI (EXTRINT, represented in ADEX at AVISIT='ALL CYCLES') is paired with the Cycle 1 ANC nadir to construct the exploratory LOESS exposure-response display (F-17-1).

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4. Efficacy Censoring Rules (ADTTE)

For Progression-Free Survival (PFS), progression is the earliest eligible, cutoff-bounded component among lesion-derived RECIST v1.0 tumour progression (OVRLRESP), reconstructed PSA progression (PSPROG), the governed F-042 pain-progression event, and death. The separately typed BSGRESP exploratory bone result and CLINPROG disposition signal are not RECIST assessments and do not feed PFS, TTUMOR, BOR, or ORR.

- Censoring Hierarchy:

1. If a patient starts a new systemic anti-cancer therapy (NACTDT) prior to a documented PFS event, the time-to-event is censored at NACTDT - 1 day (CNSDTDSC = 'NEW ANTI-CANCER THERAPY START').

2. If no event or NACT occurs, the time-to-event is censored at the latest valid post-baseline RECIST, PSA, or evaluable pain assessment (capped at the administrative cutoff). If no post-baseline assessment exists, it is censored at randomization (CNSDTDSC = 'NO POST-BASELINE ASSESSMENT').

ADTTE EVNTDESC records the typed earliest composite component: TUMOR PROGRESSION, PSA PROGRESSION, PAIN PROGRESSION, or DEATH. Same-day ties use the deterministic order tumour, PSA, pain, then death; an event on the NACT date remains an event, while strictly earlier NACT censors at NACTDT - 1. The F-042 implementation is the adopted Path A programming rule, but it remains non-sponsor-approved and requires qualified external clinical/statistical review before regulated reuse.

- Other Time-to-Event Parameters Censoring Rules (VAL-06):

- Overall Survival (OS) (PARAMCD: OS): Start date is RANDDT. Event is death (DTHFL = 'Y'). Censored at last known alive date (LSTALVDT).

- Time to First Serious AE (TTSAE) (PARAMCD: TTSAE): Start date is TRTSDT. Event is the first treatment-emergent serious AE on or before the end of safety follow-up. A non-event is censored at the earliest of LSTALVDT, TRTEDT + 30 days, and the administrative cutoff (CNSDTDSC = 'END OF SAFETY FOLLOW-UP'). (Renamed from the prior TTOS mnemonic, which was confusable with OS; the parameter is unchanged.)

- Time to PSA Progression (TTPSA) (PARAMCD: TTPSA): Start date is RANDDT. Event is post-randomization PSA progression (PARAMCD = 'PSPROG' & AVALC = 'Y'). Censored at the last eligible PSA assessment through the cutoff; if none exists, censored at randomization with CNSDTDSC = 'NO POST-BASELINE PSA ASSESSMENT'.

- Time to Tumor Progression (TTUMOR) (PARAMCD: TTUMOR): Start date is RANDDT. Event is the first post-randomization RECIST assessment with OVRLRESP = 'PD'; confirmed BSGRESP is retained as an exploratory ADRS parameter and does not feed the current TTUMOR derivation. Censored at the latest post-baseline evaluable RECIST assessment (last_tumor_dt) or randomization if none exists. Primary population is ITT (one record per ITT subject; reconstructed CbzP N=378 and real MP N=371); measurable disease (MEASDISF = 'Y') is retained as a supportive subgroup/sensitivity (CbzP N=179; MP N=203).

- Time to Pain Progression (TTPAIN) (PARAMCD: TTPAIN): Start date is RANDDT. The adopted F-042 derivation uses component-specific PPI/AS summaries, same-component confirmation at consecutive scheduled assessments at least 21 days apart, the SV visit-date hierarchy, and a direct-intent CM+PR palliative/antalgic radiotherapy union. Diary-only, RT-only, and partial/missing-date lineages are retained as supporting evidence; no terminal single-trigger exception is used. This is the restored SAP T-11-8 endpoint mapping.

4.1 Time-to-Event Analysis Conventions (audit MO-4 / MO-5)

- Duration convention. AVAL = (event/censor date - time origin) + 1 day, so an event on the origin date contributes 1 day (not 0); applied uniformly across all parameters.

- Time origin per parameter is explicit and deliberate. OS, PFS, TTPAIN, TTPSA, and TTUMOR are anchored at randomization (RANDDT, ITT); TTSAE is anchored at first dose (TRTSDT, safety). The origin is recorded on every record via STARTDT.
- Invalid durations fail the build (audit MO-4). No event or censor date is silently floored. A missing STARTDT/ADT, an ADT before STARTDT, or AVAL < 1 is a release-blocking error in both tracks. Same-day events legitimately yield one day under the +1 convention.

4A. Response Endpoint Derivations (ADRS) -- Traceability (audit F-8)

To pre-empt reviewer challenge on the response rates, the exact derivation of the response endpoints (as implemented in the SAS/R ADRS track and consumed by tfl_generation.R) is:

- Overall Response (PARAMCD = OVRLRESP) -- integrated RECIST v1.0 timepoint response. The per-visit overall response is the standard RECIST integration of three components, all sourced from ls: (1) target lesions (sum-of-diameters vs nadir/baseline, thresholds RECIST_PD_PCT/PR_PCT/PD_ABS); (2) non-target lesion status (LSCAT='NON-TARGET', worst-per-visit collapse of LSSTRESC); (3) new lesions (LSTESTCD='NEWLES'). Override rules: any new lesion => PD; non-target unequivocal PD => PD (even with target CR/PR); target CR with a non-CR non-target => PR. The derivation is defensive -- when non-target / new-lesion rows are absent for a subject-visit it reproduces the prior target-only result. The label remains "Overall Response per RECIST v1.0": new-lesion and non-target integration are part of RECIST 1.0; this is a correctness fix, not a version change. (Earlier revisions derived OVRLRESP from target SOD only and discarded the new-lesion and non-target signal that is present in the source.)
- Objective Response Rate (ORR, PARAMCD = OBJRESP): Responder = a confirmed CR or PR per RECIST v1.0 (AVALC = 'Y'). Confirmation (audit M-2) requires a subsequent CR/PR at least RECIST_CONFIRM_DAYS (28) days after the first (CR confirmed by CR; PR confirmed by CR or PR), evaluated on the lesion-derived RECIST timepoints that both the SAS and R tracks compute identically. Denominator = ITT population restricted to patients with measurable disease at baseline (MEASDISF == 'Y'), per SAP v4.0 Section 3.4 / Section 5.3 and the publication (de Bono 2010). The real MP arm yields 12/203 = 5.9% on the measurable subpopulation.
- Reconciliation to the publication: The published MP ORR was 4.4%. With confirmation enforced and overall response integrated across target + non-target + new lesions (Section 4A), the pipeline yields 5.9% on the measurable-disease denominator and 6.5% (12/185) on the response-evaluable denominator (T-11-8b). The residual reflects lesion-sum-derived RECIST versus investigator adjudication and source-date precision, not a denominator substitution.
- PSA Response (PARAMCD = PSARESP): Responder = a reconstructed >=50% confirmed decline from baseline (AVALC = 'Y') using the configured trial-analysis thresholds; this is not represented as a complete independent implementation of a later PCWG standard. The denominator is subjects with observed baseline PSA >=20 ug/L and an evaluable confirmatory PSARESP record, per SAP v4.0 Section 5.2/Table 12. Current demonstration output: CbzP 145/361 = 40.2% (synthetic) and MP 61/329 = 18.5% (real).
- Pain Response (F-042 / T-11-5): ITT subjects with PAINBL='Y' and evaluable baseline PPI/AS are assessed at the immediate next scheduled visit at least 21 days later. Response requires a component-specific PPI reduction of at least 2 points without AS worsening or an AS reduction of at least 50% without PPI worsening, with both components evaluable. The real MP arm yields 43/156 = 27.6%; the synthetic CbzP arm is reported as N/A (PN unavailable) rather than fabricated.

- Bone Scan Progression (PARAMCD = BSGRESP) -- exploratory 2+2 methodological result. New-bone-lesion rows (LSTESTCD='NEWLES' & LSLOC='BONE') are summarized separately: a first post-baseline scan with at least BONE_PROG_MIN_NEW (2) new lesions is unconfirmed progression, and a later scan adding at least BONE_PROG_CONFIRM_NEW (2) can confirm it. This later-method demonstration is not claimed as a complete PCWG3 implementation and is not in the trial-era SAP. BSGRESP is informational only: it never feeds BOR, ORR, TTUMOR, or PFS. On the real MP arm, 5 subjects reach unconfirmed progression and 0 are confirmed.

All response counts/percentages are emitted by 05_outputs/tfl/tfl_generation.R to 05_outputs/tfl/output/tables/T-11-Efficacy_Tables.txt for the current implementation. The controlled SAP-native mapping is T-11-3 PSA response, T-11-4 ORR, T-11-5 pain response, T-11-6 TTUMOR, T-11-7 TTPSA and T-11-8 TTPAIN; T-11-8b is an explicitly labelled ORR response-evaluable sensitivity. SAP v4.0 remains the authority for what is planned; generated TFLs are evidence of implementation, not the source of analysis requirements.

4B. ADAE source fidelity notes (CRF D-012 follow-on)

Fact	Evidence	Programming impact
~1,134 BASELINE AE rows have terms but blank AESER/AEREL/AEOUT/AEPATT	PDS AE extract; VISIT=BASELINE	These are non-treatment-emergent skeleton/prior-AE rows, not full on-treatment AE form completion
TEAE rows (TRTEMFL='Y')	~3,921 rows; blank AESER ~ 0-1	Safety T-20/TEAE denominators use treatment-emergent flags -- blank AESER does not dominate TEAE analyses
Grade 5	CRF grade item labels 1-4; outcome includes Fatal; AETOXGR=5 n=25 all FATAL + AESER=Y	Grade 5 is fatal-outcome toxicity mapping, not a missing CRF grade checkbox "5" on the grade line
ALB / LDH on ADSL	Not on CRF Hematology/Biochemistry panels; not in PDS LB	ALBBL/LDHBL are schema placeholders (Assigned) -- Class C for these forms, not PDS stripping of labs that were on the form

TEAE blank-AESER soft QC is logged in ADAE production/validation programs ([ADAE-QC]).

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5. Missing Data Handling (ADaMIG v1.3 Section 4.4 Compliance)

5.1 Baseline Laboratory Covariates -- Schema Placeholders (not used in any model)

Several baseline laboratory variables are carried on ADSL to satisfy the ADaM schema. Observed values are taken from deterministic source baseline records (LBBLFL='Y'); missing values remain missing. No population-median constants are inserted in the real-data track.

Variable	Real-data rule	Units	Source patient-level data available?
PSABL	Observed source baseline; otherwise missing	ug/L	Yes, where present
ALPBL	Observed source baseline; otherwise missing	U/L	Yes, where present
HGBBL	Observed source baseline; otherwise missing	g/dL	Yes, where present
ALBBL	Missing	g/L	No -- not collected; left missing (no imputation)
LDHBL	Missing	U/L	No -- not collected; left missing (no imputation)

Correction (audit closure 2026-08-09): No baseline covariate is constant-imputed in the real-data track. The primary and secondary Cox / log-rank analyses stratify only on pooled observed randomization factors (ECOGBL 0-1 vs 2 and MEASDISF; see 05_outputs/tfl/tfl_generation.R, compute_tte_stats() -> strata(ECOGBLGRP, MEASDISF)). Albumin (ALBBL) and LDH (LDHBL) are unavailable in the public MP SDTM release and remain missing with blank flags.

Flags (audit F-5). ECOGBLIF, PSABLIF, ALPBLIF, and HGBBLIF are set to 'N' because no value is imputed; they do not convert a missing observed value into an observed one. ALBBLIF and LDHBLIF are blank because the corresponding variables are unavailable and remain missing. The metadata origins and methods state this explicitly.

5.2 Analysis Window Gaps (ADLB)

The ADLB windowing schema leaves Days 35-38 unassigned (between the C2D8 window [Days 25-34] and C3D1 window [Days 39-45]). Laboratory assessments on Days 35-38 are assigned AVISITN = 99 (Unscheduled) and are excluded from the primary ANL01FL = 'Y' worst-case analysis. This is consistent with the protocol visit schedule and SAP v4.0 Section 11.1.3 (ADLB Analysis Windows -- CBC Schedule), which does not specify a Day 35-38 nominal visit.

5.3 Demographic Covariates

All subjects are assigned SEX = 'M' in A_adsl_generation.sas. This demographic assignment matches the actual study cohort (metastatic castration-resistant prostate cancer, which is exclusively male). RACE is carried from the source SDTM DM. ETHNIC was not collected in the de-identified public release; it is therefore set to the CDISC controlled-terminology value 'NOT REPORTED' and carries def:Origin Type="Assigned" (not Collected) in the define, so it is not presented as observed data. Geographic indicators COUNTRY and REGION are not present in the de-identified release and are not derived (see the note in B_bimo_generation.sas); no placeholder geography is assigned.

5.4 Analysis Populations -- Source-Inherited and Non-Discriminating (audit F-3)

The population flags ITTFL, SAFFL, and PPROTFL are carried through from the source SDTM DM (dm.itt/safety/pprot; A_adsl_generation.sas -> coalesce(...,'N'), mirrored in v_adsl_validation.R). They are not independently re-derived from inclusion/exclusion or protocol-deviation logic. Because the public de-identified PDS release is already restricted to the randomized analysis cohort, all 371 subjects carry ITTFL = SAFFL = PPROTFL = 'Y' -- the three populations coincide and are non-discriminating in this dataset. In particular, no per-protocol exclusion is exercised, because the release contains no SDTM DV (protocol-deviations) domain from which to derive one. These flags should be read as "present in the analysis cohort," and any population-based subsetting -- including the BIMO N_ITT/N_SAF/N_PPROT counts -- is a structurally-correct placeholder rather than a demonstrated filter. A production build with operational data would populate the DV domain and derive a discriminating per-protocol flag.

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6. Quality Control & SAS/R Parity (VAL-01)

6.0 Path A validation talk track (interview / reviewer)

Layer	What we do	What we do not claim
Dual-language ADaM	Independent SAS 9.4 production + R validation on real MP; cell-level recon under oda/local	Org GxP double programming (two people)
Third engine	admiral re-derivation for ADSL + OS/PFS (T1), reconciled to production	admiral for every domain
Results / figures	SAS vs R numerical recon for MP stats; forest + figure-driving gates when SAS exports exist	Pixel-identical graphics; two-arm HR as dual-programmed (CbzP synthetic)
TFL scope	Catalog-controlled in-scope set (config/tfl_output_catalog.yaml)	Full SAP Appendix D (18 IDs deferred with reasons)
Metadata	Spec->define / spec->data gates; Define XSD; local CORE ADaM rules	Commercial full Pinnacle 21 clearance
Seals	Hash-sealed Path A demo RC; scripts/verify_release.py	Part 11 e-signature / validated CSV system

Always read residuals here: WS5_KNOWN_DIFFERENCES_MEMO.md (F-003, F-005, F-011, F-012, F-014, F-025, ...).

Machine re-check (no SAS required): python3 scripts/verify_release.py

Risk tiers: RISK_BASED_VALIDATION.md config/validation_strategy.yaml

Validation is allocated by risk. Primary efficacy (OS, PFS) and ADSL get three derivation engines; supporting ADaM two; metadata automated conformance. This section documents the mechanics those tiers use.

Each ADaM dataset is produced by two independent cross-language implementations -- single-author, so this is implementation reconciliation, not two-programmer GxP double programming:

1. Production Track (SAS 9.4): modular programs under 04_analysis_datasets/programs/sas/

2. Validation Track (R 4.6.0): independent re-implementation under 04_analysis_datasets/programs/r/

Single-author disclosure (PRODUCT_CLAIM Section 2 non-claim): Both tracks were authored by one programmer. Methodological independence (different language/structure) is not organizational independence. Do not describe this package as "GxP double programming complete."

Define-XML conformance. The analysis metadata (03_metadata/define/define.xml) passes full XSD validation against the official CDISC Define-XML 2.1 + ARM v1.0 schema -- run 03_metadata/define/validate_xsd.sh (wraps xmllint against the vendored 03_metadata/define/schema/ bundle) -> "XSD: VALID." This covers the schema layer (structure, namespaces, required attributes, enumerations, element ordering) and includes Analysis Results Metadata (ARM) -- 10 ResultDisplays / 18 AnalysisResults spanning every analysis display (survival OS/PFS, secondary TTE, TEAE, OS prognostic-subgroup forest F-12, PSA waterfall F-13, exposure swimmer F-14, Project Optimus exposure-response F-17, and the lab-shift table T-21), each linking result -> method -> ADaM dataset/variables with its TFL ID named for traceability (referential integrity gated by validate_define.py). The deeper FDA/CDISC business-rule layer requires a Pinnacle 21 conformance run, and validate_define.py covers the core referential-integrity rules offline. Business-rule engine status (re-verified 2026-06-17): the open-source CDISC CORE engine (v0.16.0) was run end-to-end. CORE/CDISC Library still ships no executable ADaM rules -- directly confirmed this run: the adamig/1-0..1-3 rule sets are empty (0 rules) and update-cache fetched zero ADaM rules from the CDISC Library (only SDTMIG/SENDIG/TIG/USDM are populated, e.g. SDTMIG 3.2 = 392 rules). To obtain executable, scriptable ADaM conformance regardless, we authored ADaM rules in CORE YAML format and ran them through the real engine via --local-rules (platform/conformance_rules/adam/; latest run 7/7 SUCCESS -- see platform/conformance/CORE_RUN_RECORD.md). A CORE SDTM run was also executed (392 rules; results in platform/conformance/core_sdtm_report.json, with a documented SDTMIG 3.1.1-vs-3.2 version-gap caveat). Pinnacle 21 -- with its mature ADaM rule pack -- remains the authoritative engine for a full submission ADaM business-rule run. Both Define-XML files were also hardened to parse cleanly in the CORE reference engine (Define_XML_Version 2.1.0): the CORE run surfaced three defects the project's XSD check missed -- an invalid Role on ItemGroupDef, empty TranslatedText, and a missing def:Class element -- all fixed (def:Class added to every ItemGroupDef across both defines) while still passing XSD.

Figure terminology note (2026-06-25): F-12-1 is titled Treatment Effect Subgroup Forest Plot because the plotted quantities are within-subgroup treatment hazard ratios. The subgroup variables are prognostic factors, which explains the older ARM wording "prognostic subgroups"; that wording does not mean the figure estimates prognostic main effects.

6.1 Specification as the Single Source of Truth (audit C-4 inversion)

The authoritative analysis-dataset metadata specification is 03_metadata/adam/ADaM_spec.xlsx, authored in the CDISC / Pinnacle-21 metacore workbook format (Datasets, Variables, ValueLevel, WhereClauses, Codelists, Methods sheets). It is the metadata control source from which the rest of the metadata layer is governed -- while SAP v4.0 governs analysis intent -- reversing the previous direction, in which a reviewer workbook (ADaM_Define_Extract.xlsx) was rendered from define.xml, a circular dependency that could never disagree with the define it was meant to govern (audit finding C-4). The spec was bootstrapped once from the existing define content (one-time migration script; archived under tools/archive/migration/ -- not a live generator) and is the human-edited metadata master from then on; the old define -> extract generator is retired.

Two automated gates enforce conformance to the spec -- both run in the pipeline (cibuild Stages 27-28) and in CI:

- spec -> define (03_metadata/define/check_define_conformance.R): every dataset, variable, label, type, length, order, key sequence, mandatory flag, codelist and method in define.xml is checked against the spec; any drift fails the build. The gate ships a --self-test that injects synthetic drift and confirms detection, so it is demonstrably not a no-op. Latest run: PASS (7 datasets / 161 variables, 0 findings; platform/conformance/spec_define_conformance.json).
- spec -> data (04_analysis_datasets/programs/r/spec_data_checks.R): the produced ADaM datasets (04_analysis_datasets/adam/_prod.xpt, the SAS production track) are checked against the spec with the pharmaverse metacore + metatools + xportr toolchain -- check_variables (variable presence), check_ct_data (controlled-terminology conformance) and xportr_type/xportr_length (type/length conformance). Because the data is produced independently of the define, this is genuine (non-circular) verification. Latest run: PASS across all 7 datasets (platform/conformance/spec_data_conformance.json).

The spec also drives the variable-label artifacts applied by both tracks (platform/gen_adam_labels.R -> 04_analysis_datasets/programs/r/adam_var_labels.csv for R and 04_analysis_datasets/programs/sas/_adam_labels.sas for SAS), so production and validation carry identical, spec-sourced labels. Together these close the loop spec -> {define, data}.

SAS Execution via SAS OnDemand for Academics (ODA)

The SAS 9.4 production track (Stage 17 of the orchestrator, cibuild.py) executes on SAS OnDemand for Academics (ODA) via SASPy IOM -- a live, cloud-hosted SAS 9.4 engine (Version 9.04.01M8P022223, LIN X64) -- when the pipeline is invoked with --real-sas (ODA), or when a local SAS engine is on PATH. In those modes the SAS programs are uploaded/compiled independently and are not copied from or influenced by the R validation outputs. A verified genuine oda snapshot is retained under platform/evidence/ (byte-distinct _prod vs _v proof). Live seal mode is always read from platform/pipeline_health.json (sas_execution_mode).

Execution mode is explicit and recorded. Stage 17 resolves to exactly one of local / oda / cached / sim / error (cibuild.py -> _resolve_sas_mode) and writes the chosen mode to platform/pipeline_health.json as sas_execution_mode. Only local and oda constitute genuine, independent SAS<->R double-programming. The default invocation (python3 platform/cibuild.py with no SAS engine present) runs in sim mode -- a byte-copy of _v.xpt -> _prod.xpt -- for which a zero-difference reconciliation is tautological and is not evidence of independent parity. A reconciliation result is meaningful as double-programming evidence only for a run whose recorded sas_execution_mode is oda or local; a reviewer should confirm that field before citing the reconciliation.

The execution sequence (in oda mode) is split into two jobs through a resilient connection broker (platform/oda_broker.py); see docs/runbooks/ODA_GUIDE.md:

1. Job A -- seed (seed_sdtm.py, idempotent): the 34 SDTM SAS7BDAT files are uploaded to the ODA workspace once, guarded by a per-dataset sha256/nrows manifest (zero upload when the resident library already matches; row counts are re-read from ODA to reject a half-upload; the manifest sentinel is written last, transactionally).

2. Connect: the broker opens an IOM session with status-gated, full-jitter backoff (ODA's spawner times out under load) and earns the session via a live nonce probe -- sas_execution_mode='oda' is recorded only after the workspace echoes a runtime token.

3. Job B -- reconcile (cibuild.py --real-sas): Stage 17 verifies the SDTM manifest is resident (else it fails with an instruction to run Job A -- it does not silently simulate), uploads the SAS programs, and submits 00_master_driver.sas via %include. SAS processes the full SDTM -> Staging -> SDTM Mapping -> ADaM -> XPT chain independently.

4. The IOM log is captured to 04_analysis_datasets/programs/sas/oda_master_driver.log (WARNINGs surfaced, ERROR: fails the build), the 7 _prod.xpt are downloaded to 04_analysis_datasets/adam/, and pipeline_health.json records oda_endpoint, oda_attempts, sdtm_manifest_sha, probe_nonce_echoed, and reconciliation='SAS_vs_R'.

The cross-language reconciliation audit (Stage 18, cross_lang_audit.R) then performs an exact governed-key diffdf comparison between the independently SAS-generated _prod.xpt and the R-generated _v.xpt datasets.

The real-SAS reconciliation is a gate with teeth. On the 2026-06-18 oda run it caught a genuine SAS<->R divergence that sim mode is structurally blind to: in data work.adrs_union; set ... work.bsgresp; the AVALC length was fixed by the first contributing dataset ($20), silently truncating the PCWG3 term 'PROGRESSION UNCONFIRMED' (23 chars) to 'PROGRESSION UNCONFIR' -- flagged as 5 BSGRESP cell differences. Fixed by declaring length AVALC $100; before the SET (matching define.xml IT.ADRS.AVALC Length=100); ADRS then reconciles PASS. A sim byte-copy could never have surfaced this, because its zero-difference result is a tautology.

Results-Level Reconciliation (audit M-1, Stage 24)

Double-programming extends beyond the ADaM dataset layer to the analysis results. During the ODA run the SAS engine independently computes the MP-arm survival statistics with PROC LIFETEST (Kaplan-Meier median, event count, and N per time-to-event parameter), exported to 04_analysis_datasets/adam/tte_stats_prod.csv. Stage 24 (06_qc_evidence/reconciliation/results_reconcile.R) recomputes the identical statistics in R with survival::survfit and diffs them numerically (KM-median tolerance 1 day; event count and N exact). The verdict is written to platform/results_reconciliation_status.json and gates the build. The latest run reconciles all six parameters (OS, PFS, TTPAIN, TTPSA, TTSAE, TTUMOR) PASS.

Scope of the results reconciliation. This is the real MP arm only -- the cohort both engines derive independently. The two-arm hazard ratios and log-rank p-values shown in the TFLs are computed on MP + the synthetic CbzP comparator; because a single engine (R) holds the synthetic arm, those two-arm statistics are single-programmed by construction and are not part of the numerical SAS<->R reconciliation. In sim mode (no SAS engine) Stage 24 records not_available rather than a tautological pass.

Validation independence (audit F-1) and reconciliation scope (audit F-6). The R validation track derives every ADaM domain solely from source SDTM staging and its own logic; it does not read any _prod.xpt file. Every reconciled dataset has a governed unique business key in config/study_manifest.yaml. Occurrence-level source sequence identifiers are retained where needed (ADCM.CMSEQ, ADAE.AESEQ, ADLB.LBSEQ), and BDS keys include their parameter/visit/date components. The audit first rejects missing or duplicate keys in either track, then performs direct keyed cell comparison with diffdf. A PASS therefore certifies direct row-and-value parity between independently produced SAS and R outputs; it does not eliminate the general double-programming risk of a correlated error present in both implementations.

Decoupled MP-Only Validation Track (VAL-02)

To establish a true, functionally equivalent validation track, the core production (SAS) and validation (R) ADaM tracks process only the real Mitoxantrone (MP) safety cohort (N=371) from raw SDTM staging. The cross-language reconciliation audit (Stage 18) performs cell-by-cell diffdf verification strictly on these MP-only datasets, ensuring data structure and cell parity on the source cohort.

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7. Cabazitaxel (CbzP) Arm Reconstruction & Analysis-Step Merging

To exercise the comparative-efficacy/safety TFLs (total N=749: 371 real MP + 378 synthetic CbzP) and the retrospective Project Optimus demonstration, a synthetic, illustrative CbzP cohort was generated and merged at the analysis step. The synthetic arm is not real patient data. Two reconstruction methods are used depending on endpoint type:

- Primary endpoints (OS, PFS): Reconstructed via genuine Guyot (2012) IPD reconstruction (Guyot et al., BMC Med Res Methodol 2012;12:9), implemented with the IPDfromKM package (CRAN). The published CbzP Kaplan-Meier curves (de Bono et al., Lancet 2010;376:1147-1154, Figure 2A = OS, Figure 3 = PFS) were digitised (WebPlotDigitizer; raw exports retained) and combined with the published numbers-at-risk tables transcribed from the same figures (OS at 0/6/12/18/24/30 mo = 378/321/231/90/28/4; PFS at 0/3/6/9/12/15 mo = 378/168/90/52/15/4). The KM estimator is then inverted to solve for the event/censoring times reproducing the observed curve, constrained by N=378 and, for OS, the published death total. The CbzP survival shape comes from the published curve itself -- not from any assumed parametric form and independently of the MP arm (no hazard-ratio division) -- an accepted HTA technique (NICE TSD-14) that removes the circularity of the previous PH-scaling approach (see 01_source_data/reconstruct_cbzp_guyot.R, guyot_validation_report.R). Accuracy is bounded by digitisation fidelity, validated against the published summary statistics below.
- Secondary endpoints (TTPAIN, TTPSA, TTUMOR): Remain PH-scaled from the real MP arm (no published KM curves with numbers-at-risk tables exist for these endpoints, so Guyot reconstruction is not possible). These are clearly labelled as PH-scaled and circular in the reconstruction log and TFL footnotes.
- Non-TTE domains (ADSL, ADAE, ADEX, ADLB, ADRS): Fixed-seed sampling from published Table 1/Table 2 marginal distributions.

7.1 Separation of Reconstruction Logic

To prevent circular validation dependencies, the reconstruction program 01_source_data/reconstruct_cbzp_arm.R operates independently. For OS and PFS, it sources reconstruct_cbzp_guyot.R, which generates the Guyot pseudo-IPD from the digitised published KM curves + transcribed at-risk tables only (no MP arm data is read for the primary endpoints). For secondary endpoints, it loads the validated MP ADaM datasets (04_analysis_datasets/adam/adtte_v.xpt) to perform PH scaling. Demographics and non-TTE domains are simulated from Table 1/2 baselines. All CbzP outputs are written as isolated RDS files to 01_source_data/cbzp_reconstructed/.

7.2 Analysis-Step Merging

In the final reporting step (tfl_generation.R), the validated MP-only ADaMs are loaded from 04_analysis_datasets/adam/ and dynamically merged with the reconstructed CbzP RDS files. This combined dataset (N=749: 371 MP + 378 CbzP) is used for catalog-controlled comparative TFLs only (F-012: not protocol ITT N=755). See known-differences memo F-003 / F-012.

7.3 Demographic Reconstitution (ADSL)

Subject-level demographics for the CbzP cohort (N=378) were simulated using a fixed random seed to match baseline trial characteristics reported in Lancet 2010 Table 1:

- Age: Modeled on a normal distribution (median 68 years, range 46-92 years; ~30% < 65, ~70% >= 65).
- ECOG Performance Status: Mapped with 92% of subjects having ECOG 0-1 and 8% having ECOG 2.
- Baseline PSA: Reconstructed via a log-normal distribution matching the published median of 148 ng/mL.
- Other Stratification Factors: Prior docetaxel response (25% CR/PR), progression timeline (34% during docetaxel), measurable disease (45%), pain at baseline (59%), and visceral disease (26%).

7.4 Time-to-Event Reconstitution (ADTTE)

Primary endpoints (OS, PFS) -- Guyot (2012) IPD reconstruction:

The KM estimator is inverted (IPDfromKM) from the digitised published curve plus the transcribed numbers-at-risk table to yield pseudo-IPD consistent with the observed step function. Intrinsic reconstruction gates are blocking; comparisons against the current real-MP derivation are compatibility diagnostics because they mix a publication-derived CbzP curve with a reconstructed endpoint from the public MP source (guyot_validation_report.R):

Criterion	Published	Reconstructed	Tolerance	Status
OS median	15.1 mo (14.1-16.3)	15.2 mo	+/-1.0 mo	PASS
PFS median	2.8 mo (2.4-3.0)	2.7 mo	+/-0.5 mo	PASS
OS deaths	227 (Table 5, cabazitaxel)	228	+/-10	PASS
OS curve fit (max|dev| vs digitised)	--	0.033	<0.05	PASS
PFS curve fit (max|dev| vs digitised)	--	0.017	<0.05	PASS
OS stratified HR vs live real MP (diagnostic)	0.70 (0.59-0.83)	0.71 (0.60-0.85)	0.60-0.80	PASS
PFS stratified HR vs live real MP (diagnostic)	0.74 (0.64-0.86)	0.87 (0.75-1.02)	0.64-0.84	WARNING

The intrinsic CbzP curve-reconstruction gates pass. The live stratified OS compatibility diagnostic remains close to the published effect. The PFS diagnostic does not: the corrected real-MP endpoint uses typed RECIST/PSA/F-042 pain/death components and excludes exploratory BSGRESP/CLINPROG, producing HR 0.87. This is retained as a transparent warning and the mixed-source PFS comparison must not be described as reproducing the published effect. The paper reports no separate cabazitaxel PFS event count, so its 358 events are reconstructed from the curve and at-risk table rather than constrained to an assumed value.

Secondary endpoints (TTPAIN, TTPSA, TTUMOR) -- PH-scaled (circular):

Reconstructed using proportional-hazards scaling of the real MP event times (t_CbzP = t_MP / HR), with event counts calibrated to match published totals. These HRs are circular by construction and carry no evidentiary weight:

- Time to PSA Progression (TTPSA): PH-scaled with HR = 0.75 (286/378 events).
- Time to Tumor Progression (TTUMOR): PH-scaled with HR = 0.61 (166/378 events, ITT population; measurable-disease CbzP N=179 is retained as a supportive sensitivity).
- Time to Pain Progression (TTPAIN): PH-scaled with HR = 0.80 (130/378 events, ITT population).
- Time to Serious AE (TTSAE): Derived dynamically from the first Serious AE occurrence date in ADAE, or censored at LSTALVDT if no SAE occurred.

7.5 Adverse Events (ADAE) & Exposure (ADEX)

- Adverse Events: Simulated based on published Table 2 rates, including 82% neutropenia, 8% febrile neutropenia, 31% anemia, and 47% diarrhea. CTCAE toxicity grades and OCCDS v1.0 occurrence variables (including the custom continuous episode-merging fields CIAESDT, CIAEEDT, CIAEDUR, and occurrence flag AEOCCFL) were applied. The Serious AE (SAE) rate is calibrated to match the EPAR safety profile of exactly 39.2% (145/371 safety-evaluable subjects).
- Exposure: Simulated up to 10 cycles with standard Jevtana dosing (25 mg/m2 q3w) and cycle-level relative dose intensity (RDI) around a median of 92%, incorporating dose reductions and delays matching the publication safety profile.

7.6 Laboratory (ADLB) & Concomitant Medications (ADCM)

- Laboratory Findings: Simulated longitudinal laboratory rows (baseline and post-baseline cycles) for PSA, Haemoglobin, Platelets, and ANC. Platelet profiles and baseline-to-worst post-baseline CTCAE grade shifts are fully populated, with ~82% of patients having Grade 3/4 ANC nadirs and ~3.5% having Grade 3/4 anemia.
- Concomitant Medications: Populated with G-CSF prophylaxis usage (~8% primary, ~22% secondary prophylaxis) and post-progression starts of new anti-cancer therapies.